Mechanical Design Challenge

TDJ 2/3/4M
Technological Design
December, 2022



Mechanical Design Challenge

For this project, you will work in groups of 3 to design a machine that will accomplish a desired set of tasks.

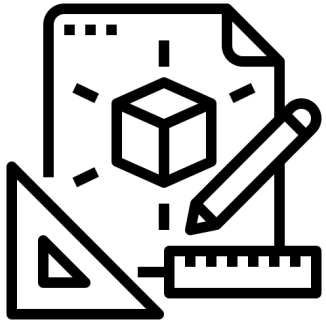
Design Criteria

1. Must lift a load of at least **200g** at least **15cm** high
2. Must weigh less than **400g**
3. **Bonus:** ability to grab, swivel, or scoop

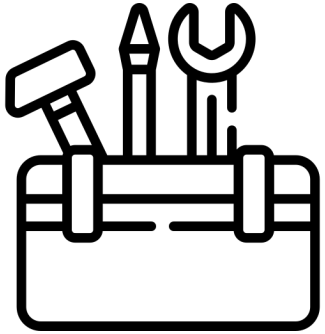
Project Roles



Research Lead



Design Lead



Construction Lead

3x long wooden sticks



60cm plastic tubing



1 x glue gun

2 x glue sticks

3x medium wooden sticks

4x wooden skewers



2× 35ml syringes



2x short wooden sticks

1x black project board

Additional Materials

1. Maximum of 2 recycled syringes
2. Unlimited amount of other recycled materials (under weight requirements)

Stage 1: Research

- Conduct research about how to build a device than can lift a load
- Research some of the materials you have access to determine how they might be used



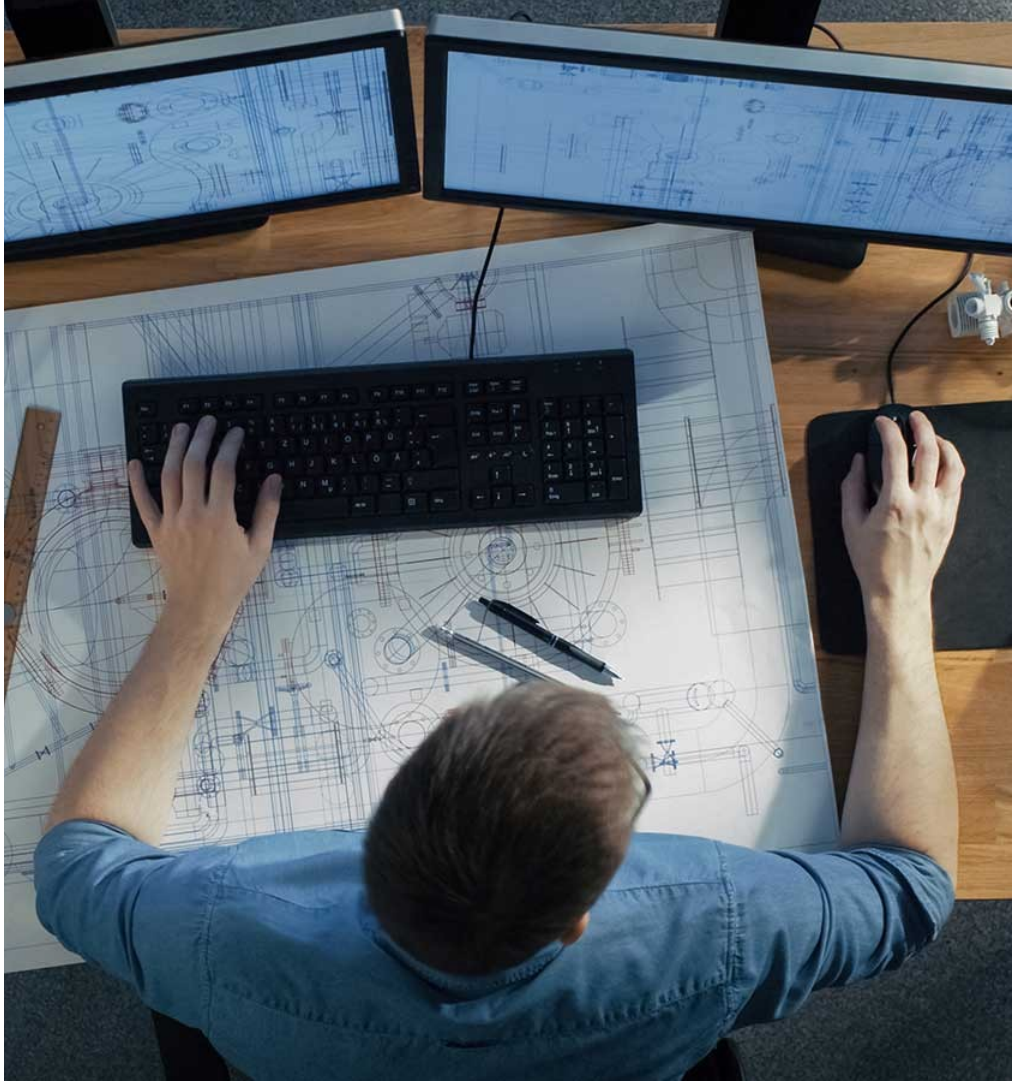
Stage 2:
Sketches

- Create concept sketches that express your design ideas
- You should include both orthographic and isometric views with projected dimensions



Stage 3:
CAD Design

- Use CAD program to bring your sketches to life
- You can use AutoCAD or SketchUp, or both
- Provide both orthographic and isometric views with dimensions
- Your design may change at this stage, and that’s OK



Stage 4:
Build & Test

- Use provided materials to build your machine based on your CAD drawings
- Test your machine and reiterate/redesign at required
- Final build will be tested in front of class for prizes



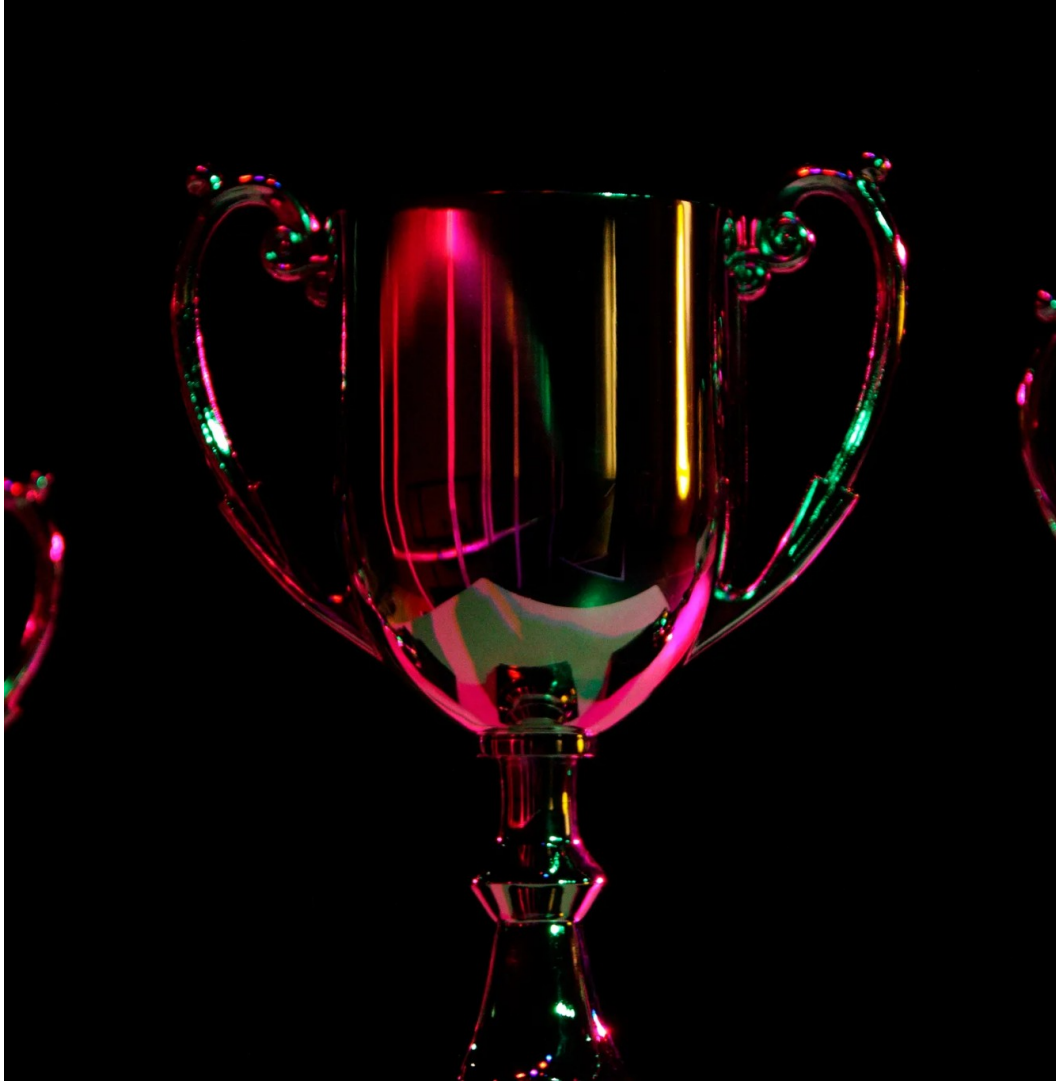
Stage 5:
Reflect

- Reflect on success of project
- Think about what you might change about your design
- Reflect on group dynamics and the extent to which everyone participated
- Self- and peer-assessment



Prize Categories

1. Most weight lifted
2. Highest load lifted
3. Largest platform
4. Most efficient
5. Most sustainable



Questions?

Use this time to clarify expectations.